

AHMED ABIDE TADESSE (Ph.D.)

Assistant Professor of Industrial Engineering &

Chair, Industrial Design and Management Science

Faculty of Mechanical and Industrial Engineering, Bahir Dar Institute of Technology (BiT)

Bahir Dar University, Ethiopia

Email: ahmed.abide@bdu.edu.et | wubied@gmail.com | **Phone:** +251923270003

PROFESSIONAL SUMMARY

A results-driven academic and researcher with unique dual Ph.D. training in Industrial Management (National Taiwan University of Science and Technology, 2023) and Industrial and Systems Engineering (Bahir Dar University, 2025). Experienced in human-robot collaboration, augmented reality, human factors engineering, and production systems. Passionate about bridging the gap between cutting-edge research and industrial applications, with a proven record of leadership, research excellence, and academic service. Currently serving as Chair of Industrial Design and Management Science under the Faculty of Mechanical and Industrial Engineering at Bahir Dar University.

EDUCATION & CERTIFICATIONS

- **Ph.D. in Industrial and Systems Engineering (Double Degree Program)**
 - *Bahir Dar University (Ethiopia)* | 2025
 - *Dissertation:* Multimodal empowerment for human robot collaboration team-a hidden semi-Markov model and augmented reality based collaborative assembly system
- **Ph.D. in Industrial Management**
 - *National Taiwan University of Science and Technology (Taiwan)* | Dec 2023
 - *Dissertation:* Multimodal empowerment for human robot collaboration team-a hidden semi-Markov model and augmented reality based collaborative assembly system

- **Higher Diploma Program (HDP) – Certified Professional Teacher Educator**
 - *Bahir Dar University (Ethiopia) | 2016*
- **M.Sc. in Production Engineering and Management**
 - *Bahir Dar University (Ethiopia) | 2015*
 - *Research Title: System Modeling and Performance Analysis Using Colored Petri Net Tools: The Case of Felege Hiwot Referral Hospital*
- **B.Sc. in Industrial Engineering**
 - *Bahir Dar University (Ethiopia) | 2010*
 - *Project Title: Analysis of facility location For ASHRAF meat processing plant*

ACADEMIC & LEADERSHIP EXPERIENCE

Bahir Dar Institute of Technology (BiT), Bahir Dar University

Chair, Industrial Design and Management Science | February 2024 – Present

- Lead academic and administrative activities, coordinating staff, curriculum development, and institutional research initiatives.
- **Accreditation Framework Adaptation:** Partnered with the ETA and BDU to adapt the national generic engineering accreditation document into a specialized program accreditation document for the B.Sc. in Industrial Engineering.
- **Curriculum Innovation:** Contributed to preparing and developing the Blended Learning Curriculum for the Industrial Management (IM) M.Sc. program.
- Spearhead the comprehensive B.Sc. in Industrial Engineering curriculum revision, successfully integrating Outcome-Based Education (OBE) frameworks.

Assistant Professor of Industrial Engineering | 2025 – Present

- Instruct undergraduate and post-graduate courses, supervise academic research, and contribute directly to departmental development.

Lecturer | 2015 – 2018

- Taught undergraduate courses, supervised research, and contributed to academic development.

Assistant Lecturer & Graduate Assistant | 2010 – 2015

- Conducted lab/tutorial sessions and assisted with foundational faculty duties.

EDITORIAL & ACADEMIC SERVICE

- **Track Chair**, International Conference on Advances of Science and Technology (ICAST), 2024.
- **Editor**, International Conference on Advances of Science and Technology (ICAST), 2024.

PUBLICATIONS

Books & Edited Volumes

1. **Metadel K A, Muluken T T, Tadesse A A, Addis A A. (2025).** *Sustainable Development Research in Green Infrastructure, Water Resources, Manufacturing, and Process Engineering: Advancements of Science and Technology*. Springer Nature.

Journal & Conference Papers

2. **Tadesse, A. A., Wang, K.-J., Lin, C. J., & Hsu, Y.-L. (2026).** A human-robot collaboration model empowered by object detection-driven AR assistance and operator behavior prediction. *International Journal of Computer Integrated Manufacturing*, 2026.
3. **Wang, K.-J., Lin, C. J., & Tadesse, A. A. (2023).** Modeling of human-robot collaboration for flexible flow shop using hidden semi-Markov models. *The International Journal of Advanced Manufacturing Technology*. Springer.
4. **Tadesse, A. A., Wang, K.-J., & Lin, C.-J. (2023).** Task allocation problem between human-robot collaboration under dynamic condition using fuzzy logic. In *Intelligent and*

Transformative Production in Pandemic Times (pp. 473-482). Springer International Publishing Cham.

5. **Lin, C.-H., Wang, K.-J., & Abide, A. T. (2022).** Human-robot collaboration empowered by hidden semi-Markov models and augmented reality. *Journal of Manufacturing Systems*, 62, 317-333. <https://doi.org/10.1016/j.jmsy.2021.12.003>
6. **Kassaneh, T. C., & Abide, A. T. (2018).** Evaluation of Workplace Environmental Ergonomics Risk Factors Using REBA and RULA Techniques. In *International Conference on Advanced Methods of Science and Technology* (pp. 102-112). Springer, Cham.
7. **Abide, A., Jayaprakash, J., & Haile, B. (2018).** Machine Repair Problem with Preventive Maintenance Strategy Using Queuing Theory. In *International Conference on Advanced Methods of Science and Technology* (pp. 83-92). Springer, Cham.

RESEARCH INTERESTS

- Human-Robot Collaboration (HRC), Object Detection, & AR-Driven Assembly Assistance
- Hidden Semi-Markov Models & Operator Behavior Prediction
- Augmented Reality (AR) & User Experience Design
- Human Factors Engineering and Workplace Ergonomics
- System Modeling and Simulation (Simio, Colored Petri Nets)
- Sustainable Development in Manufacturing and Process Engineering

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Lead Auditor Certifications** (Training completed, awaiting certificate issuance):
 - ISO 9001:2015 (Quality Management Systems)
 - ISO 14001:2015 (Environmental Management Systems)
 - ISO 45001:2018 (Occupational Health & Safety Management Systems)
- **Specialized Technical & Academic Certifications:**
 - Concentration of Data Science (July 2023)
 - Explore a Career in Machine Learning Engineering (June 2023)

- Advance Your Skills in Deep Learning and Neural Networks (March 2023)
- Getting Started with AI and Machine Learning (January 2023)
- Introduction to User Experience Design (March 2022)
- Introduction to Augmented Reality and ARCore (March 2022)
- Concentration of Human Factors Engineering (June 2020)
- Python for Machine Learning
- Mindshift: Break Through Obstacles to Learning and Discover Your Hidden Potential (February 2022)
- Learn How to Learn: Powerful Mental Tools to Help You Master Tough Subjects (February 2022)
- Work Smarter with Microsoft Excel (June 2022)

PROFESSIONAL SKILLS

- **Expert:** Technical Writing, Word Processing, Microsoft Excel
- **Intermediate:** Python, Simulation (Simio, Petri Nets), Microsoft Project

MEMBERSHIPS & LANGUAGES

- **Ethiopian Industrial Engineers Association (ESIE):** Member since July 2017
- **Languages:**
 - Amharic: Native
 - English: Expert / Fluent (15+ years' experience)
 - Arabic: Intermediate (10+ years' experience)
 - Chinese: Basic

REFERENCES

1. Prof. Kung-Jeng Wang

- Distinguished Professor of Industrial Engineering, National Taiwan University of Science and Technology
- *Role:* Doctoral Dissertation Advisor
- *Email:* kjwang@mail.ntust.edu.tw

2. **Prof. Chiuhsiang Joe Lin**

- Distinguished Professor of Industrial Engineering, National Taiwan University of Science and Technology
- *Role:* Doctoral Dissertation Advisor
- *Email:* chiuhsiangjoelin@gmail.com

3. **Dr. Bereket Haile**

- Associate Professor of Industrial Engineering, Bahir Dar University
- *Role:* Dean of Faculty of Mechanical and Industrial Engineering; Doctoral Dissertation Co-Advisor; Current Immediate Boss
- *Email:* berekethi12@gmail.com